

Leo (Yi-Ting) Chao

Irvine, CA | +1-858-519-4030 | leo.yt.chao@gmail.com | [linkedin.com/in/leoytchao](https://www.linkedin.com/in/leoytchao) | [twleochao.github.io](https://github.com/twleochao)

Education

University of California, San Diego | La Jolla, CA Sep 2026 - Present

Master of Science, Computer Science and Engineering

University of California, Irvine | Irvine, CA Sep 2023 - Jun 2026

Bachelor of Science, Computer Science - Intelligent Systems (GPA: 3.96/4.00)

Outstanding Contribution to Research (Faculty Nominated), Dean's Honors List, Accelerated 3-Year Degree

Skills

Programming Languages: Python, C++

LLM & Agents: Google ADK, Vertex AI, PyTorch, HuggingFace Transformers, LangGraph, LLM-as-a-Judge pipelines

Interpretability & Evaluation: TransformerLens, SAELens, Sparse Autoencoders (SAE), Activation Steering

Tools: GCP, Docker, Git, Linux, FastAPI

Research Experience

Julian McAuley Lab - UC San Diego | La Jolla, CA *Research Collaborator* May 2026 - Present

- Collaborating with Xin Xu on mechanistic interpretability of music LLMs using sparse autoencoders and TransformerLens; targeting NeurIPS AI4Music Workshop 2026

ARCALA Research Lab - UC Irvine | Irvine, CA *Research Assistant* Jan 2025 - Jun 2026

- Co-developed Chat-Bloom, an LLM evaluation taxonomy derived from Bloom's taxonomy, to classify and measure behavioral shifts in human-LLM interaction logs; organized and analyzed a dataset of 8,076 prompt-response pairs
- Trained BERT/DistilBERT ensemble classifiers for automated interaction taxonomy labeling, achieving ~90% accuracy validated against human-coder baseline (75.9% inter-rater reliability)
- Chao, L.**, et al. "Chat-Bloom: A Taxonomy for Classifying Cognitive Offloading in Student-AI Interactions in CS Education." Accepted, SIGCSE Virtual 2026

Professional Experience

Moovv | Irvine, CA, *AI Intern* Jun - Sep 2026

- Designed a 3-layer product-existence classifier (regex gibberish gate, web-search grounding, DeepSeek LLM judge) with a labeled eval harness; achieved 0.95 accuracy and 1.00 real-class precision/recall/F1 on a deployed consumer platform.
- Separated LLM fact extraction from deterministic lifecycle classification to prevent hallucination on unreleased products; found and fixed a Serper empty-200 reliability bug via retry logic and query reformulation

Acer | Taipei, Taiwan, *AI Systems Intern* Jun - Aug 2025

- Architected a multi-agent SDLC automation system using Google ADK and Vertex AI, reducing manual iteration time by 35%; deployed to GCP with a production path into internal Google Agentspace
- Refactored LangGraph prototype into production-grade ADK workflow with persistent memory management, integrating 6 enterprise GCP services including Cloud Storage and Firestore
- Designed state management architecture for long-horizon agent tasks, ensuring context preservation across 10+ dynamic subtask agents executing code generation, API stub creation, and test drafting

Technical Projects

LLM Activation Steering | Mechanistic Interpretability (*Python, TransformerLens, SAELens, PyTorch*) Jan - Mar 2026

- Built an inference-time intervention pipeline to inject SAE feature vectors into Gemma's residual stream, evaluating sentiment and truthfulness steering across 1,372 IMDb and 817 TruthfulQA samples
- Designed an LLM-as-a-Judge evaluation framework using GPT-5 with multi-dimensional metrics: relative sentiment shift, atomic truthfulness verification, and normalized coherence scoring
- Found myth-conviction features unexpectedly improved truthfulness scores, suggesting frequency-based selection captured entangled refusal proxies rather than pure semantic targets